

Diffusion Model for Financial Time Series Data Generation

Jason Manuel
Columbia University
116th and Broadway, New York, NY 10027
jkm2191@columbia.edu

Abstract

This project utilizes a diffusion model framework to generate synthetic financial time series data. We enhanced this framework by implementing a conditional diffusional model that took into account market regimes. Additionally, we experimented with several different models for synthetic data generation, including the Transformer, BVAE, and Perceiver architectures in tandem with a normalizing flow network. We tested our framework primarily on benchmark data, including financial indices used as reference points for the equities, fixed income, and real estate markets, improving upon the original diffusion by creating synthetic data that mimicked more closely the moments of the real data.

1. Introduction

The objective of this project was to create synthetic data that closely resembled the daily returns data for benchmark indices. Creating high quality financial synthetic data offers many opportunities from both a financial and purely machine learning perspective.

It can offer scalability as synthetic data can be generated in large volumes, allowing testing under various conditions, including stress testing, which might not be feasible with real data. Additionally, this scalability can benefit training deep neural nets when the amount of real data is incomplete or insufficient. Finally, it can also be used to augment datasets and hence improve model robustness.

2. Literature Review

We conducted a literature review of multiple papers involving the diffusion process and time series data, leveraging ideas from these papers to experiment with new model architectures for our synthesizer networks, different types of embeddings for our financial data, and also modifications to our current diffusion model.

For our synthesizer networks, we used the Bidirectional Variational Autoencoder (BVAE) [2] and the Per-

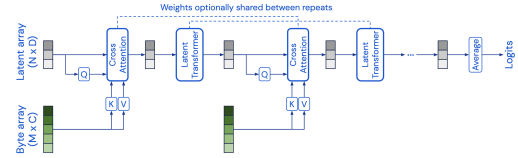


Figure 1. A diagram of the Perceiver architecture, showing the attention bottleneck and the alternation between self and cross attention

ceiver models [1]. The BVAE was inspired from a paper that sought to equip its generative time series forecasting model with high expressiveness via a coupled forward process that augments an input series and target series synchronously. However, they proposed a BVAE to take the place of the reverse process in the diffusion model. The Perceiver on the other hand was inspired from a paper that designed this model in order to handle arbitrary configurations of different modalities and which utilized a single Transformer-based architecture.

3. Methodology

The original framework which we built upon implemented a Denoising Diffusion Probabilistic Model (DDPM) [3], which gradually perturbs an original dataset with Gaussian noise while a synthesizer neural network, such as a Transformer, learns the Markov transitions of each noise addition. Then, once fully trained on the original data, the synthesizer network generates data by applying backward steps to pure Gaussian noise in order to get data that resembles the original data.

Examining the synthesizer networks, the BVAE and Perceiver architectures both look to parse the data for its key attributes by transforming it into the latent space. The Perceiver architecture expands upon this by also introducing a cross-attention module which serves as an attention bottleneck for the latent data. The architecture proceeds to iteratively apply cross-attention and self-attention, attempting to efficiently preserve the temporal trends of the data.

Model	Moment	MSE
Transformer	Mean	9.09E-08
Transformer	Standard Deviation	2.58E-07
Transformer	Skew	4.69E-01
Transformer	Kurtosis	3.87E+01
Perceiver	Mean	2.16E-07
Perceiver	Standard Deviation	1.59E-07
Perceiver	Skew	4.07E-01
Perceiver	Kurtosis	1.11E+02
BVAE	Mean	9.16E-09
BVAE	Standard Deviation	2.15E-07
BVAE	Skew	2.90E-01
BVAE	Kurtosis	9.29E+01

Figure 2. Table showing the MSE of the moments between the real and synthetic data across all model architectures

4. Experimental Results and Discussion

There are two main sets of experimental results to discuss. The ability of the synthetic data to mimic the moments of the original data, and its ability to capture the original data’s temporal dependencies.

4.1. Moments MSE

In Figure 2, we can see how each of the different synthesizer networks perform across all moments. For most moments, the Perceiver and BVAE architectures outperform the original Transformer model used previously, but struggle with kurtosis. There was no model which outperformed the others across all metrics, but the perceiver model had more training stability and could potentially outperform the other models with sufficient latent and hidden dimension size. It is also worth noting that the BVAE has the lowest MSE for mean by a large margin, suggesting that while it’s architecture is not ideal for capturing temporal dependencies, it excels at learning the mean of the original data via the latent space.

4.2. Autocorrelation

In Figure 3, we see a comparison of the autocorrelation of the real and synthetic data. Unfortunately, while we had success improving upon previous work in regards to the moments of the data, we saw little to no improvement with autocorrelation. When looking at how the total returns evolved over time between the real and synthetic data, we found that they began with positive correlation, but then this correlation quickly declined over time, as evidenced by the

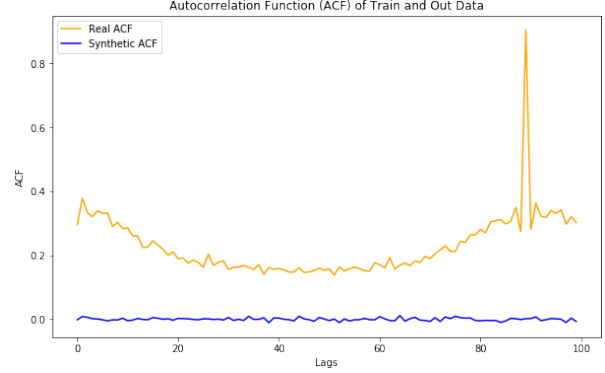


Figure 3. Graph comparing the autocorrelation between the real and synthetic data for the S&P 500 TR USD

discrepancy in autocorrelation in the graph above.

5. Conclusion

The moments of the original data were able to be closely replicated by the synthetic data generated by all the models, but especially the BVAE and Perceiver models. However, significant improvement is required in regards to capturing the temporal relationships between time series and improving the autocorrelation of the synthetic data. Future work will consider factoring in market regimes into the diffusion process so that the model can better adapt to the differences across these regimes when training the synthesizer network.

References

- [1] Andrew Jaegle, Felix Gimeno, Andrew Brock, Andrew Zisserman, Oriol Vinyals, and Joao Carreira. Perceiver: General perception with iterative attention, 2021. <https://arxiv.org/pdf/2103.03206>. 1
- [2] Yan Li, Xinjiang Lu, Yaqing Wang, and Dejing Dou. Generative time series forecasting with diffusion, denoise, and disentanglement, 2022. 1
- [3] Timur Sattarov, Marco Schreyer, and Damian Borth. Findiff: Diffusion models for financial tabular data generation, 2023. <https://arxiv.org/pdf/2309.01472>. 1